

MACHINE LEARNING FUNDAMENTALS - LS2026

SEMINAR: KERNEL FUNCTIONS

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Assignment 1. Consider a feature map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^{d(d+1)/2}$ whose entries are

$$\phi(\mathbf{x}) = \begin{pmatrix} x_1^2, & \sqrt{2}x_1x_2, & \sqrt{2}x_1x_3, & \dots, & \sqrt{2}x_1x_d, \\ & x_2^2, & \sqrt{2}x_2x_3, & \dots, & \sqrt{2}x_2x_d, \\ & & \vdots & & \\ & & & & x_d^2 \end{pmatrix}^T,$$

so that the features correspond to all possible products of unordered pairs of entries from $\mathbf{x}$, and the products of different entries are multiplied by a constant factor $\sqrt{2}$. For example, if $\mathbf{x} = (x_1, x_2, x_3)^T \in \mathbb{R}^3$ then

$$\phi(\mathbf{x}) = (x_1^2, \sqrt{2}x_1x_2, \sqrt{2}x_1x_3, x_2^2, \sqrt{2}x_2x_3, x_3^2)^T.$$

This feature map defines a kernel $k(\mathbf{x}, \mathbf{x}') = \langle \phi'(\mathbf{x}), \phi'(\mathbf{x}') \rangle$ referred to as the homogeneous polynomial kernel of degree 2. Show that the kernel value equals to the square of the dot product of the input vectors, that is prove the identity

$$k(\mathbf{x}, \mathbf{x}') = \langle \phi'(\mathbf{x}), \phi'(\mathbf{x}') \rangle = \langle \mathbf{x}, \mathbf{x}' \rangle^2, \quad \forall \mathbf{x}, \mathbf{x}' \in \mathbb{R}^d.$$

Assignment 2. Given a training set $T_m = ((\mathbf{x}_i, y_i) \in \mathbb{R}^d \times \mathbb{R} \mid i = 1, \dots, m)$, consider the linear regressor

$$h(\mathbf{x}; \mathbf{w}, b) = \langle \mathbf{w}, \mathbf{x} \rangle + b,$$

whose parameters $(\mathbf{w}, b) \in \mathbb{R}^{d+1}$ are learned from T_m by solving the least-squares problem. The resulting parameters are given by

$$\begin{bmatrix} \mathbf{w}_m \\ b_m \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y},$$

where

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top & 1 \\ \vdots & \vdots \\ \mathbf{x}_m^\top & 1 \end{bmatrix} \in \mathbb{R}^{m \times (d+1)}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \in \mathbb{R}^m.$$

Show that linear regression can be expressed solely in terms of dot products between input vectors. In particular, derive a formulation of the predictor that depends only on inner products

$$\langle \mathbf{x}_i, \mathbf{x}_j \rangle,$$

and explain how this leads to a kernelized version in which these dot products are replaced by a kernel

$$k(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle.$$